RHINOQUANT

RESEARCH / MATERIALS INTELLIGENCE

STRATEGIC MATERIALS | UK

The UK Subsurface Materials Opportunity Map

Aluminium / Magnesium / Helium
and selected strategic by-products

SCREENING CONCLUSION

Process first. Mine selectively.

Best near-term route: existing dolomite / industrial feedstocks; Antrim bauxite is niche; helium requires existing-well data first.

RESEARCH PROPOSAL

13 October 2026 | Desk-based assessment

INVESTMENT THESIS

A selective, evidence-led UK materials strategy

RhinoQuant should not begin with a greenfield mine thesis. The public evidence supports a portfolio built around **industrial mineral upgrading, by-product recovery and targeted subsurface screening** - with exploration capital released only when assays, recoveries, logistics and offtake justify it. [1-4]

895,166 t	70,317 t	3,359 t	Mg-first
Aluminium imports in 2025	Magnesium imports in 2025	Helium imports in 2025	Near-term feedstock route

Priority screen (RhinoQuant analyst judgement; not a reserve classification)

Opportunity	Evidence today	Commercial case	Proposed decision
Magnesium from existing dolomite supply	Strong occurrence / established quarry districts [1]	Medium for Mg compounds; low-to-medium for primary metal	**Priority 1** - qualify product/feed routes
Antrim bauxite / aluminium-bearing clays	Historic mining and documented occurrences; variable quality [2]	Low for bulk alumina; niche uses remain untested	**Priority 2** - test specialty applications only
Helium in legacy gas systems	Trace helium historically reported in some North Sea fields [5,6]	Unproven; concentration and recoverable volume unknown	**Priority 3** - data-room screen, no blind drilling
Ga / Ge / In and silica / strontium by-products	Regional geological indicators, but no assured deposit [1]	Option value if attached to another viable feedstock	**Watchlist** - sample only in shared campaigns

Core investment view. Commercialising already mined or disturbed material is a more credible first step than opening a new mine. The value proposition is secure UK feedstock, reliable processing, and qualified material for industrial customers - not an assumption that geology alone creates a profitable asset.

The proposal is consistent with the UK's 2025 Critical Minerals Strategy / Vision 2035 direction, which recognises aluminium, magnesium, helium and several related elements as critical or growth minerals. [3]

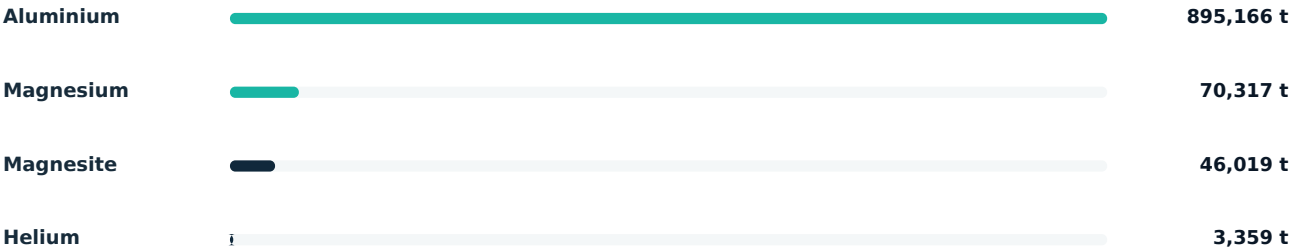
MARKET SIGNAL

Imports are the commercial signal; geology is only half the equation

HMRC-derived figures published in a UK Parliamentary answer show material import volumes for aluminium, magnesium, magnesite and helium in 2025. These are **trade volumes, not a direct measure of domestic consumption**; product definitions and re-exports can affect the relationship. [4]

Material	EU-origin imports	Non-EU imports	Total 2025	Non-EU share
Aluminium	468,440 t	426,726 t	895,166 t	47.7%
Magnesium	33,385 t	36,932 t	70,317 t	52.5%
Magnesite	24,737 t	21,282 t	46,019 t	46.2%
Helium	740 t	2,619 t	3,359 t	78.0%

Imported material volumes (2025)



Demand outlook: government growth-sector estimates are explicitly modelled as **cumulative** demand signals, not annual consumption. The 2025 technical annex estimates cumulative demand of about 8.0 Mt aluminium and 44,750 t helium for relevant Industrial Strategy growth sectors by 2035. They signal future strategic importance, but should not be compared directly to the import totals above. [3]

Commercial implication. Prioritise customer discovery and imported-product specification analysis before geological spend: alloys and shapes for aluminium; MgO / refractory / chemical grades for magnesium; high-purity gas and supply-assurance contracts for helium.

COMMODITY 01

Aluminium: a real occurrence, a weak bulk-ore thesis

The clearest documented UK bauxite occurrences are in County Antrim, Northern Ireland. They formed as lateritic weathering horizons within the Palaeogene Antrim Lava Group. The BGS account describes a variable residual clay containing kaolin-group minerals, gibbsite and boehmite alongside quartz, goethite and minor titanium minerals. [2]

Historic extraction is meaningful geological evidence but not a modern reserve statement. The BGS account records approximately 244,000 tonnes mined during the Second World War and notes that individual samples exceeded 62% Al_2O_3 with low silica and iron oxides, while other material was sub-ore grade. The same source concludes that Antrim deposits are not competitive with low-cost high-grade imports for bulk bauxite; it notes a small quantity reported at Clinty near Ballymena for ferrous aluminium sulphate water-treatment feedstock. Current activity and permissions require verification. [2]

RhinoQuant interpretation

Do not underwrite a primary aluminium mine or alumina refinery on public data. The most defensible path is a small programme testing whether selective horizons or legacy / disturbed material can command a premium in chemical, ceramic, refractory, adsorbent or water-treatment applications. The adjacent gallium angle is geologically interesting in theory, but the national prospectivity assessment says UK bauxite is very limited and Antrim bauxitic layers are not substantial enough to be considered a potential mineral deposit for that purpose. [1]

Question	Screening test	Go / no-go logic
Feed quality	Representative channel / bulk samples; XRF/ICP chemistry; mineralogy; loss on ignition	Stop if high silica / iron or grade variability destroys recovery or customer specification
Specialty product fit	Bench test calcination, acid solubility, particle size, whiteness / impurity limits where relevant	Proceed only with a named buyer specification and repeatable product
Resource & permissions	Historical plans, land ownership, planning history, protected-area constraints, tonnage reconciliation	No resource statement without survey, sampling design and competent-person review
Logistics & residue	Haul distance, water, energy, waste and rehabilitation costs	Compare delivered cost against imported substitute, not against mine-gate price

Recommendation: *an 8-12 week specialty-materials test is justified only if local access and a credible end-user exist. Treat bulk alumina economics as a negative base case and any niche product as an unproven option. [2]*

COMMODITY 02

Magnesium: leverage established dolomite before pursuing metal

The UK has extensive magnesium-bearing carbonate geology. The BGS / UKCMIC national assessment identifies dolomite in the Durness Group of north-west Scotland; dolomitised Appin Group limestones in the Central Highlands and Islay; Permian Raisby, Ford, Cadeby and Brotherton formations in north-east England; and Carboniferous Monsal Dale Limestone in Derbyshire, the Pembroke Limestone Group in south Wales and the Black Rock Limestone group in the south Wales-Bristol area. [1]

The same assessment identifies carnallite associated with deep Permian Zechstein evaporites beneath east Yorkshire and continuing offshore. Magnesite / brucite occurrences in ultramafic and serpentinite rocks are irregular and were not considered robust targets in that national assessment; olivine-bearing rocks in western Scotland and Northern Ireland were also not judged a potential resource in that work. [1]

The important commercial distinction is between **dolomite as an industrial mineral**, magnesium compounds such as magnesia / hydroxide, and primary magnesium metal. Existing dolomite quarrying gives a practical feedstock and infrastructure route, but dolomite contains calcium as well as magnesium; converting it to high-purity MgO or Mg metal adds separation, energy and process-cost burdens. A mineable carbonate body is not automatically an economic magnesium-metal project.

Priority regions / routes

Target area	Why it matters	Recommended first action
North-east England	Permian dolomite formations; established industrial / transport context	Map operating and dormant quarries, chemistry, reserves and nearby industrial consumers
Derbyshire	Monsal Dale Limestone and a mature industrial-mineral region	Check quality, conservation/planning designations and co-product / residue routes
South Wales - Bristol	Pembroke Limestone and Black Rock Limestone groups	Screen steel, refractory, cement, water-treatment and chemical users
North-west Scotland / Islay	Older dolomite / dolomitised limestone occurrences	Regional geological interest; logistics and environmental constraints may dominate
East Yorkshire deep evaporites	Carnallite-bearing Zechstein sequence at depth	Data review only; not a near-term hard-rock mining target

Preferred pilot thesis: recover value from existing dolomite, fines or industrial residues into a defined magnesium compound or engineered mineral product. Run a parallel engineering study for low-carbon magnesium extraction only after feed assays, reagent consumption, energy intensity, impurity removal and buyer requirements are known. The project should compete on delivered product quality and carbon / energy intensity, not on the idea that magnesium-bearing rock is scarce. [1,7]

COMMODITY 03

Helium: high strategic value, unproven UK deposit economics

Helium is included on the UK Critical Minerals List and the 2025 growth-minerals list. The government's demand model flags medical MRI and semiconductor uses as important drivers. Parliamentary trade data show 3,359 tonnes of helium imports in 2025, of which 2,619 tonnes were recorded as non-EU origin. The government's modelled cumulative growth-sector demand signal reaches 44,750 tonnes by 2035; this is a scenario-based cumulative figure, not an annual demand forecast. [3,4]

There is historical evidence of helium in the UK North Sea petroleum system, but not proof of a commercially recoverable helium field. In a 1971 Parliamentary answer, West Sole, Leman Bank, Indefatigable, Viking and Rough were reported to contain a *very small amount* of helium. A 1985 study found North Sea gas-field helium to be dominantly radiogenic, consistent with crustal uranium / thorium decay and fault-related transport. These sources are historical and do not provide modern concentrations, recoverable volumes or economic studies. [5,6]

RhinoQuant recommendation: data first, no blind helium drilling

Stage	Work	Release criterion
1. Data-room screen	Request historic gas compositions / chromatograms from UK Continental Shelf and onshore well owners; review well logs, reservoir data, field life and processing trains	Prioritise only assets with measured He values, accessible samples and adequate gas-flow / reservoir context
2. Assay campaign	Reanalyse archived gas samples or collect approved samples; independent lab assays for He and relevant impurities	Replicate assays and material concentration above a buyer- and process-specific threshold
3. Recovery concept	Model gas throughput, recovery rate, purification, compression, energy, coproducts and transport; interview specialty-gas offtakers	Positive contribution margin under downside pricing / recovery scenarios and credible offtake
4. Pilot / field work	Only after stages 1-3; partner with licence holder / operator and regulators	No capital commitment without access rights, permitting pathway and independently reviewed reservoir case

What must be true: measurable helium concentration; sufficient gas in place and deliverability; a route to separate helium without compromising other products; available processing / power; and a customer willing to contract for the specification. The most plausible near-term project is a **screening and coproduct-recovery study across existing wells**, not greenfield resource speculation.

Current evidence rating: low confidence in commerciality. No commercially proven UK helium reserve or production case is asserted in this proposal.

ADJACENT UPSIDE

Screen by-products and high-purity feedstocks, not every mapped mineral

A selective watchlist can improve project returns where it shares sampling, infrastructure or processing with the primary opportunity. The UKCMIC/BGS assessment is explicit that its maps identify geological potential, not guaranteed deposits, and do not by themselves resolve environmental, planning or economic constraints. [1]

Material	UK evidence / prospective setting	RhinoQuant screen
Gallium (Ga), germanium (Ge), indium (In)	Potential as by-products in UK zinc, copper or tin mineral systems; very limited bauxite and no substantial Antrim bauxite deposit identified for Ga. [1]	Add assays to polymetallic mineral samples and relevant process residues. Do not finance a standalone mine without a primary-metal business case.
High-purity silica / silicon feed	The BGS assessment says no vein-quartz deposit of sufficient size for silicon-metal feed was identified; three quartzite formations (Scaraben, Loch Treig, Holyhead) had limited analyses at 98-99% SiO2, with resource potential uncertain. [1]	Only progress if bulk quality, impurity profile, scale, energy and customer spec clear high-purity economics.
Strontium / celestine	Historic extraction from celestine-rich nodules and veins in Triassic Mercia Mudstone around Bristol, south-west England; Somerset-Gloucester-Monmouth region flagged as potentially prospective. [1]	GIS and legacy-mine / borehole data review; sample only when host mineral and target market justify it.
Secondary / urban mining	Aluminium scrap, magnesium-containing process residues, spent refractories and mineral by-products can reduce dependence on new ore, although quality and recovery vary. [3,7]	Run a feedstock inventory alongside geology. Prioritise clean, consistent and traceable streams that already sit near industrial demand.

Illustrative ranking - based on access to existing material / infrastructure, not on theoretical geological potential

Rank	Workstream	Rationale
1	Dolomite upgrading / magnesium compounds	Known feedstocks and existing industrial-mineral activity; fastest route to testable samples and customers.
2	Antrim bauxite specialty applications	Documented geology and legacy extraction, but bulk-ore economics are weak.
3	Helium in existing well / gas data	High strategic value; geologically plausible but commercial concentration and volumes are unknown.
4	Ga / Ge / In, silica, strontium	Portfolio options only; piggyback on other sampling and process streams.

A project should not advance solely because an area is labelled “prospective”. Require measurable grade / concentration, recoverability, continuity / throughput, access, permitting and a customer-specification route.

PROPOSED MANDATE

A staged 12-month validation programme

RhinoQuant should operate as the investment and commercial-screening lead, supported by a competent geologist, industrial mineral specialist, analytical laboratory, reservoir / gas-processing engineer, and planning / land-access adviser. Release funds at stage gates; do not bundle later field cost into the first cheque.

Phase	Timing	Scope	Indicative budget*
A. Evidence and GIS	Weeks 1-8	BGS / UKCMIC data, BritPits, planning and environmental overlays, historical mine plans, import / customer map, well data requests, target ranking	£35k-£60k
B. Sample and specification	Months 2-5	Representative available samples; XRF / ICP / mineralogy; bauxite calcination / chemical tests; dolomite / Mg process screening; gas re-assays where lawful and sample access exists; buyer interviews	£75k-£180k
C. Field validation	Months 5-12	Site access, field mapping, confirmatory sampling, limited geophysics where it answers a defined question, logistics / water / energy studies, conceptual flowsheets and environmental baseline	£150k-£400k
D. Pilot / drilling decision	After month 12	Independent resource / reservoir review; pilot design and customer qualification. Drilling, major plant and land acquisition budgeted separately.	Separate approval

Indicative planning estimates prepared for this proposal, not vendor quotations. Overall A-C envelope: approximately* **£260k-£640k*. Excludes deep drilling, major plant, land acquisition, licensing transactions, VAT and unusual access / remediation costs. Scope may reduce materially if archived samples and well data are readily available.*

Decision gates and deliverables

Gate	Deliverable	Pass condition
G1 - target confirmation	Ranked GIS target register, land / planning screen, existing data inventory, customer map	At least one accessible feed / data set and a defined product or recoverable-gas hypothesis
G2 - technical fit	Assay pack, mineralogy / gas results, initial process routes, buyer spec comparison	Repeatable feed quality and credible recovery to a specification; no fatal impurity / logistics issue
G3 - economics	Downside / base / upside unit economics, capex ranges, energy / water / residue accounting, offtake letters	Positive margin at conservative prices and adequate process / resource life
G4 - investment committee	Independent technical review and development recommendation	Proceed, partner, defer or stop - documented with explicit uncertainties

Expected result after Phase C: not an investment bankable feasibility study, but a credible decision package identifying whether any asset warrants a technical-economic study, pilot or licence-holder partnership.

RISK, GOVERNANCE & SOURCES

The proposal is a screening mandate, not a reserve claim

The evidence base is strongest for mineral occurrence and strategic demand, and weakest for deposit-scale tonnage, representative grades, recoveries, capex and achievable sales prices. Those gaps must remain visible in all investment materials.

Risk	Control
Geology / grade risk	Use representative sampling plans, duplicates / blanks / standards, QA/QC and independent review; never extrapolate isolated high-grade samples to a resource.
Market / product risk	Secure buyer specifications and indicative offtake before process optimisation; compare delivered cost and full impurity profile against imports.
Process / energy risk	Quantify yield, reagents, heat, electricity, water, emissions and residues; stress-test UK power and carbon-cost scenarios.
Planning / ESG / access	Overlay protected sites, groundwater, landscape, biodiversity, heritage, land ownership and community impacts at the start; do not assume a historic mine can reopen.
Helium data risk	Modern, representative gas assays and reservoir / flow data are mandatory; historical trace evidence alone is not a business case.
Governance risk	Use stage gates, documented kill criteria, independent technical review, and no public reserve / resource language without a recognised reporting standard and competent person.

Sources and key reference points

[1] Deady, E. et al. (2023), *Potential for Critical Raw Material Prospectivity in the UK*, British Geological Survey, CR/23/024. National-scale geological screening; magnesium, Ga/Ge/In, silicon, strontium and assessment limitations. https://www.ukcmic.org/downloads/reports/ukcmic-potential-for-critical-raw-material-prospectivity-in-the-uk-cr23024.pdf

[2] British Geological Survey / Earthwise, *Bulk minerals, mineral resources, Northern Ireland*, bauxite section (Antrim geology, historical production, grade variability and competition). https://earthwise.bgs.ac.uk/index.php/Bulk_minerals,_mineral_resources,_Northern_Ireland

[3] Department for Business and Trade, *Critical Minerals technical annex*, updated 23 January 2026; critical / growth mineral lists and cumulative demand methodology. https://www.gov.uk/government/publications/uk-critical-minerals-strategy/critical-minerals-technical-annex

[4] UK Parliament, Written Question 7959, answer 12 June 2026; HMRC-derived import volumes by EU / non-EU origin for 2023-2025. https://questions-statements.parliament.uk/written-questions/detail/2026-06-08/7959

[5] House of Commons, Hansard, “North Sea Gas”, 24 May 1971; historical answer lists five UK-sector gas fields said to contain very small amounts of helium. https://api.parliament.uk/historic-hansard/written-answers/1971/may/24/north-sea-gas

[6] Hooker, P., O’Nions, R. & Oxburgh, E. (1985), “Helium isotopes in North Sea gas fields and the Rhine rift”, *Nature* 318, 273-275. Geological background, not modern commercial resource evidence. https://www.nature.com/articles/318273a0

[7] British Geological Survey, *United Kingdom Minerals Yearbook 2025*, production, consumption, trade and value series; revised data note dated 29 July 2026. https://www.bgs.ac.uk/mineralsuk/

[8] British Geological Survey, *Mineral resources* dataset and planning maps; national mapped distribution is not an economic reserve statement. https://www.bgs.ac.uk/datasets/bgs-mineral-resources/

RhinoQuant conclusion. Start with industrial magnesium feedstock qualification and a tightly scoped Antrim specialty-material test. Run a low-cost helium data-room screen in parallel, but release no field or drilling capital until contemporary assays and reservoir economics support it. Carry Ga/Ge/In, silica and strontium only as by-product options. The objective is a staged route from UK geological evidence to qualified product or recoverable gas - with clear stop conditions at every stage.

Important limitation: this is desk research and a proposed work programme. It does not estimate a JORC / NI 43-101 / PERC mineral resource or reserve, guarantee a deposit, constitute engineering feasibility, or provide investment advice. All budget figures are indicative and require supplier quotations.